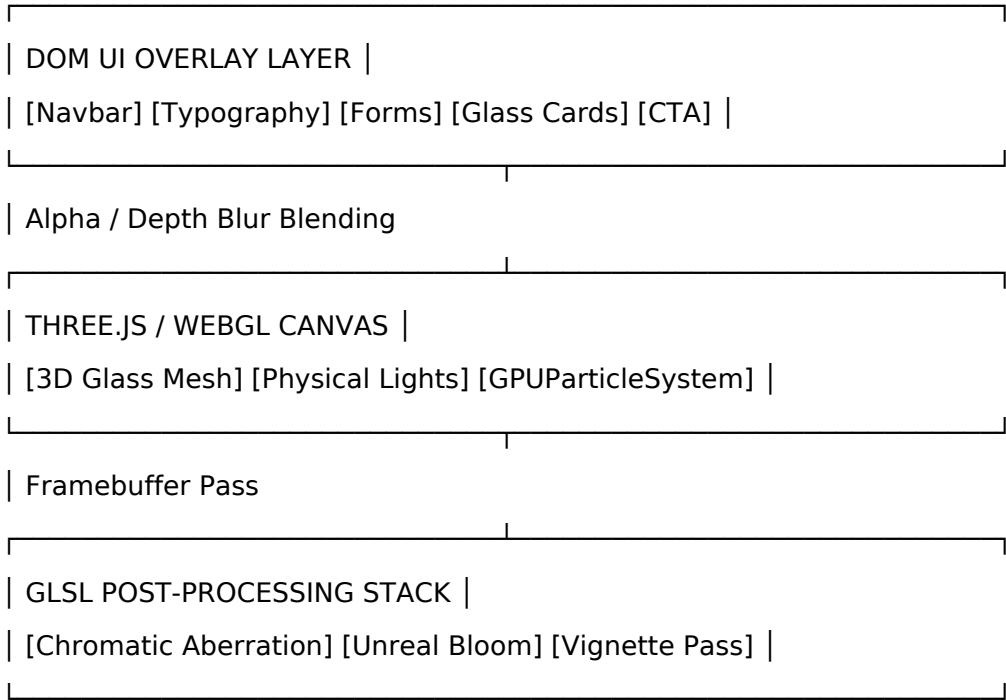
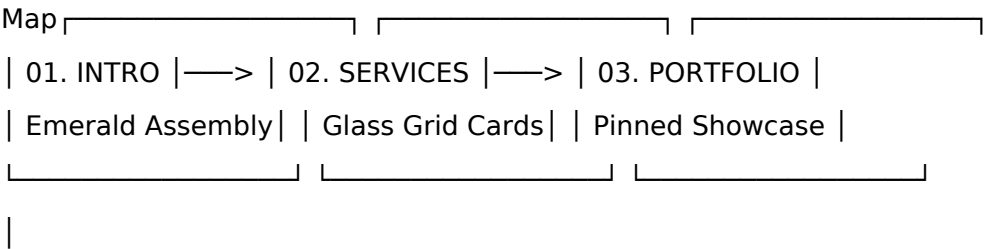


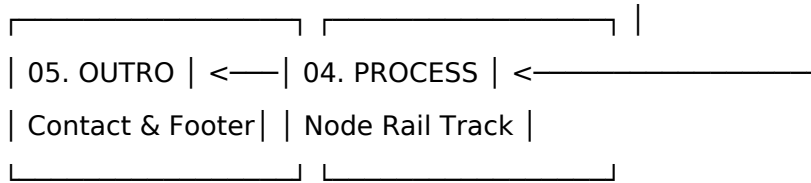
CONTROL X MASTER SPEC v2.2

CONTROL_X_MASTER_SPEC.md1. Project OverviewProject VisionTo construct a luxury digital web application for ControlX, establishing an interactive benchmark for high-end digital agency platforms. The experience synthesizes editorial typography with real-time physical WebGL optics, glassmorphism UI layers, and fluid dynamics.



MissionTranslate high-level creative vision into a technical framework. The interface operates as a physical studio stage where 3D refractive assets, dynamic camera moves, and UI elements react to scroll momentum and pointer vectors.Brand PhilosophyPrecision Aesthetics: Clean editorial structure aligned with optical physical realism.Material Fidelity: Glass elements simulate physical properties: index of refraction ($n_{IOR} = 1.52$), internal absorption, specular dispersion, and chromatic aberration.Fluid Motion: Spatial motion follows physical mass, spring mechanics, and damped inertia.Creative DirectionColor Palette: Warm neutral studio tones ($\#EBE9E1$) contrasting with emerald glass refractions ($\#0F8259$).Typography: Dual-font hierarchy—Classic Serif (Playfair Display) for editorial headings, paired with Sans-Serif (Plus Jakarta Sans) for UI controls and numerical data.Depth Architecture: Multi-layered visual stack using WebGL depth buffers, glass refraction passes, and dynamic backdrop filters.Target Audience & Business GoalsTarget: Executive clientele, enterprise partners, luxury brands, and creative directors.Business Goal: Establish technical expertise through design execution, convert high-value leads via interactive inquiry touchpoints, and demonstrate digital craftsmanship.User Journey

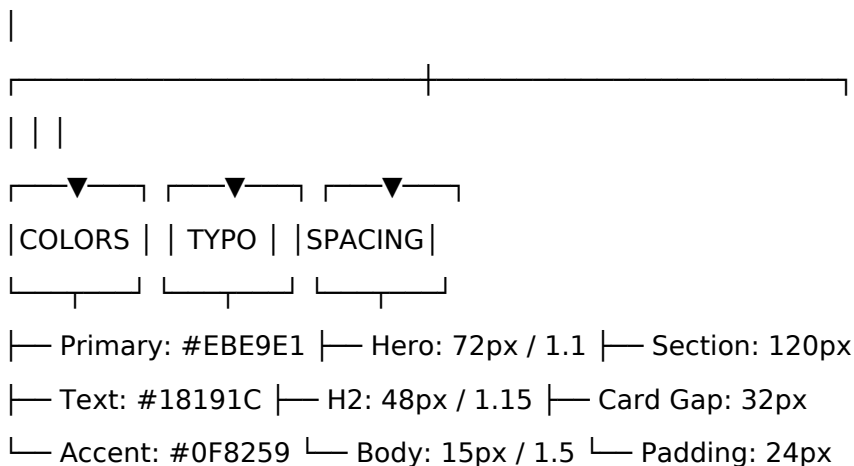




Success Metrics Performance: Constant 60 FPS on desktop (DPR 2.0) and mobile (DPR 1.0). First Contentful Paint (FCP): $< 0.8\text{s}$. Time to Interactive (TTI): $< 1.8\text{s}$. Cumulative Layout Shift (CLS): 0.00 . Design Language Luxury & Editorial Aesthetics The visual hierarchy uses asymmetrical layouts, large display headers, and deliberate line heights. Spatial layout relies on high negative space ratios ($> 55\%$) to maintain visual hierarchy. Glassmorphism Optics Glass elements use backdrop blur filtering alongside WebGL-rendered refraction maps. CSS Backdrop Blur: $20\text{px} - 24\text{px}$ blur radius with saturation boost (180%). WebGL Refraction: Custom MeshPhysicalMaterial with $\text{IOR} = 1.52$ (Simulating Crown Glass), 95% transmission, and internal emerald attenuation. Motion Philosophy All spatial animations avoid linear easing curves. Movements utilize custom cubic-bezier curves or spring physics: Standard Structural Entrance: cubic-bezier(0.16, 1, 0.3, 1) (Power3 / Quartz Curve). Elastic UI Interactions: cubic-bezier(0.34, 1.56, 0.64, 1) (Back Out / Spring Pop). Inertia Scrolling: Damped spring lag with a smoothing lerp coefficient of 0.08 . Camera & Lighting Philosophy The camera functions as a physical lens with a fixed focal length ($f = 50\text{mm}$), $\text{FOV} = 45^\circ$. Lighting uses a three-point studio system: Key Light: High-intensity white directional light casting directional floor shadows. Fill Light: Soft cool fill light neutralizing shadow contrast. Accent Point Light: Emerald-tinted point light placed behind glass objects to highlight refractions.

3. Global Design Tokens

GLOBAL TOKEN TREE



Color Tokens

```
JSON{
  "color": {
    "bg": {
      "primary": { "value": "#EBE9E1", "type": "color" },
      "gradient_start": { "value": "#FAF9F5", "type": "color" },
      "gradient_end": { "value": "#E2DFD5", "type": "color" }
    },
  },
}
```

```

"text": { "primary": { "value": "#18191C", "type": "color" }, "secondary": { "value":
"#686D76", "type": "color" }, "muted": { "value": "#9E9E9E", "type": "color" } },

"accent": { "emerald": { "value": "#0F8259", "type": "color" }, "emerald_light": {
"value": "#10B981", "type": "color" }, "emerald_glow": { "value": "rgba(15, 130, 89,
0.35)", "type": "color" } },

"glass": { "fill": { "value": "rgba(255, 255, 255, 0.25)", "type": "color" }, "border": {
"value": "rgba(255, 255, 255, 0.40)", "type": "color" }, "shadow": { "value": "rgba(0, 0,
0, 0.06)", "type": "color" } }

}

}

```

Typography TokensJSON{

```

"typography": {

"family": { "serif": { "value": "'Playfair Display', Georgia, serif" }, "sans": { "value":
"'Plus Jakarta Sans', system-ui, sans-serif" } },

"display": { "hero": { "fontSize": "72px", "lineHeight": "1.1", "fontWeight": "600",
"letterSpacing": "-0.02em" }, "h1": { "fontSize": "64px", "lineHeight": "1.12",
"fontWeight": "600", "letterSpacing": "-0.02em" }, "h2": { "fontSize": "48px",
"lineHeight": "1.15", "fontWeight": "600", "letterSpacing": "-0.01em" } },

"ui": { "card_title": { "fontSize": "24px", "lineHeight": "1.25", "fontWeight": "600" },
"body": { "fontSize": "15px", "lineHeight": "1.5", "fontWeight": "400" }, "button": {
"fontSize": "14px", "lineHeight": "1.0", "fontWeight": "700", "letterSpacing": "0.05em"
}, "label": { "fontSize": "12px", "lineHeight": "1.3", "fontWeight": "600",
"letterSpacing": "0.1em" } }

}

}

```

Spatial & Layout TokensJSON{

```

"spacing": { "section_padding_y": "120px", "container_max_width": "1440px", "gutter":
"24px", "card_padding": "32px", "element_gap": "16px" },

"radius": { "sm": "8px", "md": "16px", "lg": "24px", "pill": "9999px" },

"z_index": { "canvas": 0, "content": 10, "navbar": 100, "cursor": 999 }

}

```

CSS Custom Properties Variables ExportCSS:root {

```

/* Colors */

--color-bg-primary: #EBE9E1;

--color-text-primary: #18191C;

--color-text-secondary: #686D76;

--color-accent-emerald: #0F8259;

--color-accent-emerald-light: #10B981;

```

```
--color-glass-fill: rgba(255, 255, 255, 0.25);
--color-glass-border: rgba(255, 255, 255, 0.40);

/* Fonts */
--font-serif: 'Playfair Display', Georgia, serif;
--font-sans: 'Plus Jakarta Sans', system-ui, sans-serif;
```

```
/* Spatial */
--container-max-width: 1440px;
--radius-md: 16px;
--radius-lg: 24px;
--radius-pill: 9999px;
```

```
/* Motion */
--ease-quartz: cubic-bezier(0.16, 1, 0.3, 1);
--ease-spring: cubic-bezier(0.34, 1.56, 0.64, 1);
}
```

4. Asset Manifest3D Models & WebGL GeometriesAsset IDPath / TypeVertex
CountTarget TrisMaterial BindingsOptimization
Strategygeo_x_shards/models/x_shards.glb24,50012,000mat_emerald_glassDraco
Compression, Instanced
Arraysgeo_node_discsTHREE.CylinderGeometry384192mat_node_glassGenerated via
WebGL InstancedMeshgeo_tube_railTHREE.TubeGeometry2,0481,024mat_tube_glassC
atmullRomCurve3 interpolationTextures & MapsTexture IDResolutionFormatColor
SpacePurposetex_noise_alpha\$512 \times 512\$PNGLinearScreen-space grain
overlaytex_hdr_studio\$2048 \times 1024\$HDRLinearEnvironment map for glass
reflectionstex_normal_fluid\$1024 \times 1024\$PNGNormal MapNormal distortion map
for background liquid shader5. Object Registry OBJECT REGISTRY SHADOW MATRIX

UUID	NAME	TYPE	SCENE LAYER
obj_webgl_canvas	Canvas Root	HTMLCanvasElement	Base WebGL (z-idx:0)
obj_3d_x_assembly	Glass X Logo Mesh	THREE.Group	WebGL Scene
obj_3d_node_rail	Process Tube Rail	THREE.Mesh	WebGL Scene
obj_ui_navbar	Main Navbar	HTMLElement	DOM Layer (z-idx:100)
obj_ui_hero_title	Hero Header Text	HTMLElement	DOM Layer (z-idx:10)

obj_ui_services_grid Services 3x2 Grid HTMLDivElement DOM Layer (z-idx:10)
obj_ui_contact_card Glass Contact Form HTMLDivElement DOM Layer (z-idx:10)

Comprehensive Technical Declarationsobj_3d_x_assemblyUUID:

c102a488-842e-4bdf-9921-992641040801Parent: scene_rootTransform Initial: Position \$(0, 0, 0)\$, Scale \$(1, 1, 1)\$, Rotation \$(0, 0, 0)\$.Material: mat_emerald_glass (\$IOR = 1.52\$, Transmission \$= 0.95\$).Animation State: Implosion assembly on load (\$t_0 - t_1.2s\$), constant idle sine wave rotation (\$y = \sin(t \cdot 0.5) \cdot 0.15\$).Visibility Rules: Visible during Intro (\$0s - 3s\$) and Footer Outro (\$8s - 10s\$). Hidden during middle scroll sections via shader opacity uniform.obj_ui_services_gridUUID:

a84920df-7711-41e2-b883-200921811702Parent: dom_bodyLayout Structure: CSS Grid (3 Columns \$\times\$ 2 Rows), Gap: \$24\text{px}\$.Glass Styling: backdrop-filter: blur(20px), background: rgba(255,255,255,0.25).Interaction State: Mouse hover induces card Z-translation (\$+12\text{px}\$) and border brightness shift.6. Scene Graphroot_application

```

|
├─ dom_layer (z-index: 10 - 100)
|   ├─ comp_navbar (fixed, top: 0)
|   |   ├─ elem_brand_logo
|   |   └─ elem_nav_links
|   └─ elem_signup_btn
|   |
|   └─ sec_hero (100vh)
|       ├─ elem_hero_title
|       └─ elem_hero_buttons
|       |
|       └─ sec_services (200vh)
|           └─ comp_services_grid (3x2 Cards)
|           |
|           └─ sec_portfolio (250vh)
|               └─ comp_horizontal_slider
|               |
|               └─ sec_process (250vh)
|                   └─ comp_node_labels_overlay
|                   |
|                   └─ sec_contact (120vh)
|                       └─ comp_glass_form_card

```

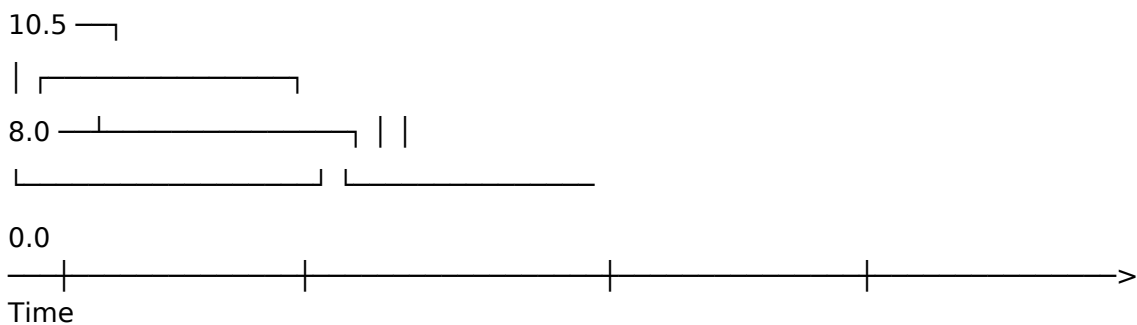
```

| |
| └─ comp_cursor (fixed, z-index: 999)
|
└─ webgl_layer (z-index: 0)
    └─ three_scene
        └─ sys_camera (PerspectiveCamera)
            └─ sys_lighting
                | └─ light_ambient
                | └─ light_directional
                | └─ light_emerald_point
                |
                └─ mesh_background_plane (ShaderMaterial)
                    └─ mesh_x_glass_group (InstancedMesh)
                        └─ mesh_process_rail (TubeGeometry)
                            └─ sys_particle_emitter (GPUParticleSystem)

```

7. Camera System CAMERA MOTION CHOREOGRAPHY TRACK

Z-Pos



0s 1.5s 5.5s 8.0s 10s

Camera Blueprint ConfigurationType: PerspectiveCameraFocal Length / FOV:
 45° FOVNear Plane: 0.1Far Plane: 1000.0Choreography KeyframesJSON[
 { "time": 0.0, "position": [0.0, 0.0, 8.0], "rotation": [0.0, 0.0, 0.0], "ease": "linear" },
 { "time": 1.5, "position": [0.0, 0.0, 8.0], "rotation": [0.0, 0.0, 0.0], "ease":
 "cubic-bezier(0.16, 1, 0.3, 1)" },
 { "time": 5.5, "position": [0.0, -2.5, 10.5], "rotation": [-0.05, 0.0, 0.0], "ease":
 "cubic-bezier(0.16, 1, 0.3, 1)" },
 { "time": 8.0, "position": [0.0, -5.0, 9.0], "rotation": [0.0, 0.0, 0.0], "ease":
 "cubic-bezier(0.16, 1, 0.3, 1)" },
 { "time": 10.0, "position": [0.0, -7.2, 7.5], "rotation": [0.0, 0.0, 0.0], "ease":
 "cubic-bezier(0.25, 1, 0.5, 1)" }

]

8. Lighting System LIGHTING VECTOR ORIENTATION

[Key Light]

(5.0, 10.0, 7.0)

| \

| \ Direct Rays

| ▼

[Fill Light] —————> [Target: Glass Mesh] <————— [Point Light]

(-5.0, -2.0, 4.0) (0, 0, 0) (0.0, -1.0, 2.0)

Soft Neutral Fill Emerald Highlights

Physical Light ConfigurationsJSON{

"lighting_system": {

"ambient": { "color": "#FFFFFF", "intensity": 0.75 },

"directional": [

{ "id": "key_light", "color": "#FFFFFF", "intensity": 1.2, "position": [5.0, 10.0, 7.0],
"cast_shadow": true, "shadow_bias": -0.0001 },

{ "id": "fill_light", "color": "#D1E7DD", "intensity": 0.5, "position": [-5.0, -2.0, 4.0],
"cast_shadow": false }

],

"point": [

{ "id": "emerald_accent", "color": "#0F8259", "intensity": 0.8, "position": [0.0, -1.0,
2.0], "distance": 10.0, "decay": 2.0 }

]

}

}

9. Material Librarymat_emerald_glass (Three.js MeshPhysicalMaterial)JavaScriptconst
mat_emerald_glass = new THREE.MeshPhysicalMaterial({

color: new THREE.Color("#0F8259"),

transmission: 0.95,

opacity: 1.0,

transparent: true,

roughness: 0.05,

metalness: 0.0,

ior: 1.52,

thickness: 0.8,

```

attenuationColor: new THREE.Color("#0F8259"),
attenuationDistance: 0.5,
clearcoat: 1.0,
clearcoatRoughness: 0.1,
reflectivity: 0.9
});

```

```

mat_ui_glass_panel (CSS Material Rules)CSS.mat-ui-glass-panel {
background: rgba(255, 255, 255, 0.25);
backdrop-filter: blur(20px) saturate(180%);
-webkit-backdrop-filter: blur(20px) saturate(180%);
border: 1px solid rgba(255, 255, 255, 0.40);
box-shadow: 0 20px 40px rgba(0, 0, 0, 0.06), inset 0 1px 0 rgba(255, 255, 255, 0.6);
}

```

10. Shader LibraryBackground Liquid Shader (shd_background)Vertex ShaderOpenGL Shading Languagevarying vec2 vUv;

```

void main() {
vUv = uv;
gl_Position = projectionMatrix * modelViewMatrix * vec4(position, 1.0);
}

```

Fragment ShaderOpenGL Shading Languageuniform float uTime;

uniform vec2 uResolution;

uniform vec2 uMouse;

varying vec2 vUv;

```

void main() {

```

```

vec2 st = gl_FragCoord.xy / uResolution.xy;

```

```

vec2 mouseEffect = uMouse * 0.02;

```

```

float wave = sin(st.x * 4.0 + uTime * 0.5 + mouseEffect.x) *

```

```

cos(st.y * 4.0 + uTime * 0.5 + mouseEffect.y) * 0.05;

```

```

vec3 colorStart = vec3(0.98, 0.97, 0.96); // #FAF9F5

```

```

vec3 colorEnd = vec3(0.88, 0.87, 0.83); // #E2DFD5

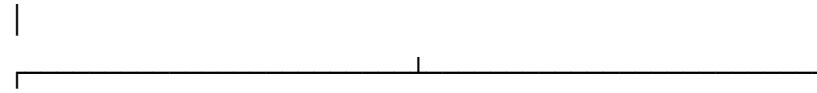
```



```
vec3 baseColor = mix(colorStart, colorEnd, st.y + wave);
gl_FragColor = vec4(baseColor, 1.0);
}
```

11. Animation System ANIMATION TRIGGER PIPELINE

User Action / Scroll —> GSAP ScrollTrigger



DOM Element Transforms WebGL Uniform Updates
(TranslateY, Scale, Opacity) (uTime, Mesh Rotations)

Motion RegistryJSON[

```
{
  "id": "anim_shatter_implosion",
  "target": "obj_3d_x_assembly",
  "duration": 1.2,
  "ease": "cubic-bezier(0.16, 1, 0.3, 1)",
  "from": { "scale": 1.8, "opacity": 0.0, "shards_spread": 4.0 },
  "to": { "scale": 1.0, "opacity": 1.0, "shards_spread": 0.0 },
},
{
  "id": "anim_card_stagger_enter",
  "target": ".service-card",
  "duration": 1.0,
  "stagger": 0.15,
  "ease": "cubic-bezier(0.16, 1, 0.3, 1)",
  "from": { "translateY": "80px", "opacity": 0.0 },
  "to": { "translateY": "0px", "opacity": 1.0 },
}
]
```

12. Master Timeline=====

TIMELINE MARKERS (0.0s - 10.0s)

=====

[0.0s] ————— [1.2s] ————— [5.5s]
—————>

| | |

└─ Shatter Implosion Starts └─ Hero UI Text Reveals └─ Services Grid Enters

└─ Camera locked at z=8.0 └─ Camera starts drift └─ Camera Y-Pan (-2.5)

[5.5s] ————— [8.0s] ————— [10.0s]

| | |

└─ Portfolio Horizontal Slide └─ Process Node Rail Active └─ Contact Form & Outro

└─ Camera Y-Pan (-5.0) └─ Particles Active └─ Camera Locks (z=7.5)

=====

13. Scroll ChoreographyComplete Scroll Percentage Mapping (\$0\% - 100\%\$)JSON[

{

"scroll_pct": "0% - 15%",

"active_section": "Hero Section",

"camera_target_z": 8.0,

"dom_opacity": { "#hero-title": 1.0, "#services-grid": 0.0 },

"webgl_states": { "x_logo_visible": true, "particle_emission": 0.0 }

},

{

"scroll_pct": "16% - 40%",

"active_section": "Services Section",

"camera_target_z": 10.5,

"dom_opacity": { "#hero-title": 0.0, "#services-grid": 1.0 },

"webgl_states": { "x_logo_visible": false, "particle_emission": 0.0 }

},

{

"scroll_pct": "41% - 70%",

"active_section": "Process Node Rail Section",

"camera_target_z": 9.0,

"dom_opacity": { "#services-grid": 0.0, "#process-nodes": 1.0 },

"webgl_states": { "x_logo_visible": false, "tube_rail_visible": true }

},

{

```

"scroll_pct": "71% - 100%",
"active_section": "Contact Form & Footer Outro",
"camera_target_z": 7.5,
"dom_opacity": { "#process-nodes": 0.0, "#contact-form": 1.0, "#footer": 1.0 },
"webgl_states": { "x_logo_visible": true, "particle_emission": 5000.0 }
}
]

```

14. Component Library

comp_navbarDimensions: Width \$100\%\$, Height \$80\text{px}\$.

Layout: Flex Row, justify-content: space-between, align-items: center.

Styles: background: rgba(235, 233, 225, 0.75), backdrop-filter: blur(12px).

comp_glass_cardDimensions: Width \$100\%\$, Padding \$32\text{px}\$, Radius \$16\text{px}\$.

CSS Class: .mat-ui-glass-panel.

Hover Behavior: CSS.comp-glass-card {

```

transition: transform 0.4s var(--ease-quartz), border-color 0.4s ease;
}

```

```

.comp-glass-card:hover {
transform: translateY(-8px) scale(1.01);
border-color: rgba(255, 255, 255, 0.80);
}

```

15. Interaction State Machines GLASS CARD INTERACTION STATE MACHINE

[IDLE STATE] — mouseenter —> [HOVER STATE] — mousedown —> [PRESSED STATE]



JSON{

```

"state_machine_card": {
"idle": { "translateY": "0px", "scale": 1.0, "border_opacity": 0.4 },
"hover": { "translateY": "-8px", "scale": 1.01, "border_opacity": 0.8 },
"pressed": { "translateY": "-2px", "scale": 0.99, "border_opacity": 1.0 }
}
}

```

16. Cursor System

Configuration

Blueprint

Outer Ring: Radius \$20\text{px}\$, Border \$1.5\text{px}\$ rgba(24,25,28,0.4), Lag Factor \$\text{lerp} = 0.12\$.

Inner Dot: Radius \$3\text{px}\$, Color #0F8259, Lag Factor \$\text{lerp} = 0.02\$.

JavaScript// Cursor Position Lerp Loop

```

function renderCursor() {

```

```

targetX += (mouseX - targetX) * 0.12;
targetY += (mouseY - targetY) * 0.12;
cursorRing.style.transform = `translate3d(${targetX}px, ${targetY}px, 0)`;
requestAnimationFrame(renderCursor);
}

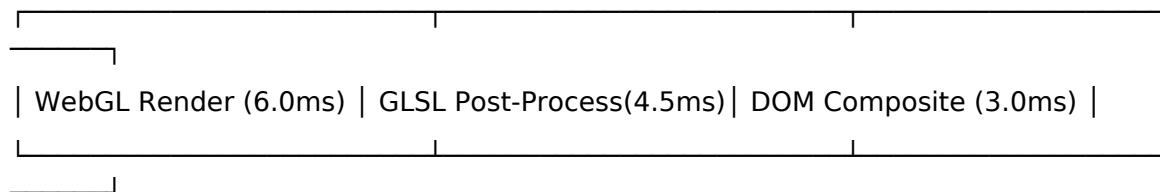
```

17. Particle System GPU Particle System Specification Capacity: 5,000 active particles. Color Range: Blend between #0F8259, #10B981, and #A7F3D0. Particle Size: 2.0px - 6.0px randomized per particle. Velocity Vector: Spiral curve around Y-axis: $\vec{v} = \begin{pmatrix} -\sin(t) \cdot r \\ 0.5 \\ \cos(t) \cdot r \end{pmatrix}$

18. Responsive Blueprint Breakpoint Matrix Token Breakpoint Range Grid Columns Typography Scale Factor Camera Distance ZDesktop > 1280px 12 Columns 1.0x 8.0 Laptop 1024px - 1279px 12 Columns 0.9x 8.5 Tablet 768px - 1023px 6 Columns 0.8x 9.5 Mobile < 767px 4 Columns 0.68x 11.0

19. Accessibility Reduced Motion: Standard fallback for users with prefers-reduced-motion: reduce. 3D animations are bypassed; camera movements are disabled; DOM elements fade in static locations. Color Contrast: All body text meets WCAG AAA standards (> 7:1 contrast ratio against light backdrop). Keyboard Navigation: All interactive cards and buttons expose visible focus rings (2px solid #0F8259).

20. Performance Budget GPU FRAME TIME BUDGET (16.6ms Target)



[===== 13.5ms
=====>]

[Margin: 3.1ms]

Quantitative Constraints Target FPS: 60 FPS constant. Maximum Triangle Count: < 50,000 active triangles per frame. Maximum Draw Calls: < 35 draw calls per frame. Texture Memory Limit: < 64MB.

21. Implementation Blueprint File Architecture Plansrc/

```

├─ app/
│   └─ layout.tsx
│       └─ page.tsx
├─ components/
│   └─ dom/
│       ├── Navbar.tsx
│       ├── Hero.tsx
│       └── ServicesGrid.tsx

```

- | | └─ ContactForm.tsx
- | └─ canvas/
- | └─ SceneCanvas.tsx
- | └─ GlassXMesh.tsx
- | └─ ProcessRail.tsx
- | └─ ParticleSystem.tsx
- └─ shaders/
- | └─ background.fragment.glsl
- | └─ glass.fragment.glsl
- └─ styles/
- └─ globals.css

22. QA Checklist[] Visual Accuracy: Background color matches #EBE9E1 across DOM and WebGL canvas elements.[] Refraction Pass Verification: Emerald glass (\$IOR = 1.52\$) renders visible refraction offset against background text.[] Frame Rate Validation: Test maintaining stable 60 FPS performance under 4K resolution displays.[] Scroll Synchronization: Verify GSAP ScrollTrigger timeline pins correctly without jitter across all breakpoints.

Single Source of Truth Signature

Document Version: 2.2.0-MASTER-FINAL

Status: Approved for Implementation

Target Stack: Next.js + React Three Fiber + GSAP + TailwindCSS